SSL Certificate and DNS Setup

Last updated by | Haroun Ahmady | Apr 14, 2026 at 9:08 AM PDT

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Overview

This guide covers:

- Obtaining SSL/TLS certificates
- DNS record configuration
- Certificate installation in Nginx
- Certificate renewal procedures

DNS Configuration

Create DNS Record

Work with your DNS administrator to create the following record:

Type	Name	Value	TTL
A	clientiq.fmb.com	10.x.x.x (server IP)	300

For high availability with multiple servers:

Type	Name	Value	TTL
A	clientiq.fmb.com	10.x.x.1	300
A	clientiq.fmb.com	10.x.x.2	300

Or use a load balancer:

Type	Name	Value	TTL
CNAME	clientiq.fmb.com	lb-clientiq.fmb.com	300

Verify DNS Resolution

```
# Windows
nslookup clientiq.fmb.com
Resolve-DnsName clientiq.fmb.com

# Expected output:
# Name:    clientiq.fmb.com
# Address: 10.x.x.x
```

Certificate Options

Option 1: Internal Certificate Authority (Recommended for Enterprise)

Most banks have an internal PKI. Request a certificate from your security team:

Certificate Request Information:

- Common Name (CN): `clientiq.fmb.com`
- Subject Alternative Names (SAN): `clientiq.fmb.com`
- Key Size: 2048-bit or 4096-bit RSA
- Signature Algorithm: SHA-256
- Validity: 1-2 years
- Key Usage: Digital Signature, Key Encipherment
- Extended Key Usage: Server Authentication

Option 2: Generate CSR (Certificate Signing Request)

If you need to generate a CSR:

```powershell
# Create OpenSSL config file
@"
[req]
default_bits = 2048
prompt = no
default_md = sha256
distinguished_name = dn
req_extensions = req_ext

[dn]
C = US
ST = California
L = San Francisco
O = Your Bank Name
OU = Information Technology
CN = clientiq.fmb.com

[req_ext]
subjectAltName = @alt_names

[alt_names]
DNS.1 = clientiq.fmb.com
"@ | Out-File -FilePath "C:\nginx\conf\ssl\openssl.cnf" -Encoding ASCII

# Generate private key and CSR
cd C:\nginx\conf\ssl
openssl req -new -newkey rsa:2048 -nodes `
    -keyout clientiq.key `
    -out clientiq.csr `
    -config openssl.cnf

# View CSR content (to submit to CA)
openssl req -in clientiq.csr -noout -text
```

Note: If OpenSSL is not available, download from https://slproweb.com/products/Win32OpenSSL.html

Option 3: Wildcard Certificate

If your organization has a wildcard certificate (e.g., `*.fmb.com`):

1. Obtain the certificate and key from your security team
2. The certificate should cover `*.fmb.com`
3. Copy to `C:\nginx\conf\ssl\`

Certificate Installation

Step 1: Prepare Certificate Files

You should have received:

- **Certificate file** (`.crt` or `.cer`): The signed certificate
- **Private key** (`.key`): Generated during CSR creation
- **CA Bundle** (optional): Intermediate certificates

Place files in `C:\nginx\conf\ssl\` :

```
C:\nginx\conf\ssl\
├── clientiq.crt         # Your certificate
├── clientiq.key         # Private key
└── ca-bundle.crt        # Intermediate certificates (if provided)
```

Step 2: Create Certificate Chain (if needed)

If you received separate intermediate certificates:

```
# Combine certificate with CA bundle
Get-Content "C:\nginx\conf\ssl\clientiq.crt", "C:\nginx\conf\ssl\ca-bundle.crt" |
    Set-Content "C:\nginx\conf\ssl\clientiq-chain.crt"
```

Update `nginx.conf` to use the chain:

```
ssl_certificate     ssl/clientiq-chain.crt;
ssl_certificate_key ssl/clientiq.key;
```

Step 3: Set File Permissions

```
# Secure the private key
$acl = Get-Acl "C:\nginx\conf\ssl\clientiq.key"
$acl.SetAccessRuleProtection($true, $false)

# Only SYSTEM and Administrators
$rule = New-Object System.Security.AccessControl.FileSystemAccessRule(
    "SYSTEM", "Read", "None", "None", "Allow"
)
$acl.SetAccessRule($rule)
$rule = New-Object System.Security.AccessControl.FileSystemAccessRule(
    "Administrators", "FullControl", "None", "None", "Allow"
)
$acl.SetAccessRule($rule)

Set-Acl "C:\nginx\conf\ssl\clientiq.key" $acl
```

Step 4: Verify Certificate

```
# Check certificate details
openssl x509 -in "C:\nginx\conf\ssl\clientiq.crt" -noout -text

# Verify key matches certificate
$certMod = openssl x509 -noout -modulus -in "C:\nginx\conf\ssl\clientiq.crt" | openssl md5
$keyMod = openssl rsa -noout -modulus -in "C:\nginx\conf\ssl\clientiq.key" | openssl md5

if ($certMod -eq $keyMod) {
    Write-Host "Certificate and key match!" -ForegroundColor Green
} else {
    Write-Host "ERROR: Certificate and key do NOT match!" -ForegroundColor Red
}

# Check certificate expiration
openssl x509 -in "C:\nginx\conf\ssl\clientiq.crt" -noout -dates
```

Step 5: Test Nginx Configuration

```
cd C:\nginx
.\nginx.exe -t
# Expected: nginx: configuration file C:\nginx/conf/nginx.conf test is successful
```

Step 6: Reload Nginx

```
.\nginx.exe -s reload
# or
Restart-Service Nginx
```

Verify SSL Configuration

Test with Browser

1. Navigate to `https://clientiq.fmb.com`
2. Click the padlock icon
3. Verify:
   - Certificate is valid
   - Issued to: clientiq.fmb.com
   - Issued by: Your CA
   - Expiration date is correct

Test with PowerShell

```
# Basic HTTPS test
$response = Invoke-WebRequest -Uri "https://clientiq.fmb.com/health" -UseBasicParsing
$response.StatusCode

# Check SSL certificate
$uri = [System.Uri]"https://clientiq.fmb.com"
$request = [System.Net.HttpWebRequest]::Create($uri)
$request.ServicePoint.Expect100Continue = $false
try {
    $response = $request.GetResponse()
    $cert = $request.ServicePoint.Certificate

    Write-Host "Subject: $($cert.Subject)"
    Write-Host "Issuer: $($cert.Issuer)"
    Write-Host "Valid From: $($cert.GetEffectiveDateString())"
    Write-Host "Valid To: $($cert.GetExpirationDateString())"
} finally {
    if ($response) { $response.Close() }
}
```

Test with OpenSSL

```
# Check full certificate chain
openssl s_client -connect clientiq.fmb.com:443 -servername clientiq.fmb.com

# Check specific TLS versions
openssl s_client -connect clientiq.fmb.com:443 -tls1_2
openssl s_client -connect clientiq.fmb.com:443 -tls1_3
```

Online SSL Test

Use SSL Labs for comprehensive testing (if externally accessible): https://www.ssllabs.com/ssltest/

Certificate Renewal

Monitoring Certificate Expiration

Create a monitoring script `C:\nginx\scripts\check-cert.ps1` :

```
$cert = New-Object System.Security.Cryptography.X509Certificates.X509Certificate2(
    "C:\nginx\conf\ssl\clientiq.crt"
)

$daysUntilExpiry = ($cert.NotAfter - (Get-Date)).Days

if ($daysUntilExpiry -lt 30) {
    $subject = "URGENT: ClientIQ SSL Certificate Expiring in $daysUntilExpiry days"
    $body = @"
The SSL certificate for ClientIQ will expire on $($cert.NotAfter).

Server: $env:COMPUTERNAME
Certificate Subject: $($cert.Subject)
Certificate Issuer: $($cert.Issuer)
Expiration Date: $($cert.NotAfter)
Days Until Expiry: $daysUntilExpiry

Please renew the certificate immediately.
"@

    # Send email alert (configure SMTP settings)
    # Send-MailMessage -To "ops@fmb.com" -From "alerts@fmb.com" `
    #     -Subject $subject -Body $body -SmtpServer "smtp.fmb.com"

    Write-Host $subject -ForegroundColor Red
    Write-EventLog -LogName Application -Source "ClientIQ" -EventId 1001 `
        -EntryType Warning -Message $body
} else {
    Write-Host "Certificate valid for $daysUntilExpiry days" -ForegroundColor Green
}
```

Schedule daily check:

```
$action = New-ScheduledTaskAction -Execute "PowerShell.exe" `
    -Argument "-NoProfile -ExecutionPolicy Bypass -File C:\nginx\scripts\check-cert.ps1"
$trigger = New-ScheduledTaskTrigger -Daily -At "8:00AM"
$principal = New-ScheduledTaskPrincipal -UserId "SYSTEM" -RunLevel Highest

Register-ScheduledTask -TaskName "ClientIQ-CertCheck" -Action $action -Trigger $trigger -Principal $principal
```

Renewal Process

1. **30 days before expiry**: Generate new CSR (if key rotation required)
2. **14 days before expiry**: Submit CSR to CA
3. **7 days before expiry**: Install new certificate
4. **Day of install**:

   ```powershell
   # Backup old certificate
   Copy-Item "C:\nginx\conf\ssl\clientiq.crt" "C:\nginx\conf\ssl\clientiq.crt.backup"

   # Install new certificate
   Copy-Item "C:\Temp\new-clientiq.crt" "C:\nginx\conf\ssl\clientiq.crt"

   # Test and reload
   cd C:\nginx
   .\nginx.exe -t
   .\nginx.exe -s reload

   # Verify
   openssl x509 -in "C:\nginx\conf\ssl\clientiq.crt" -noout -dates
   ```

Troubleshooting

Certificate Not Trusted

1. Check if CA certificate is in Windows trust store
2. Verify intermediate certificates are included in chain
3. Test with: `openssl verify -CAfile ca-bundle.crt clientiq.crt`

Certificate/Key Mismatch

```powershell
# Compare modulus hashes
openssl x509 -noout -modulus -in clientiq.crt | openssl md5
openssl rsa -noout -modulus -in clientiq.key | openssl md5
# These should match
```

Nginx Won't Start with SSL

1. Check key file permissions
2. Verify key is not encrypted:

   ```powershell
   openssl rsa -in clientiq.key -check
   # Should output "RSA key ok" without password prompt
   ```

3. If key is encrypted, decrypt it:

   ```powershell
   openssl rsa -in clientiq.key -out clientiq-decrypted.key
   ```

Mixed Content Warnings

Ensure all resources load over HTTPS. Check for:

- Hard-coded HTTP URLs in code

- External scripts/styles over HTTP
- API calls to HTTP endpoints

Security Best Practices

1. **Use strong key sizes**: Minimum 2048-bit RSA
2. **Enable only TLS 1.2+**: Disable SSLv3, TLS 1.0, TLS 1.1
3. **Use strong cipher suites**: Prefer ECDHE and GCM
4. **Enable HSTS** (after testing): Prevents downgrade attacks
5. **Implement OCSP stapling**: Faster certificate validation
6. **Rotate keys on renewal**: Generate new key pair each time
7. **Monitor expiration**: Automated alerts 30+ days before expiry

Next Steps

1. Configure SAML Authentication
2. Review Environment Variables